Evaluation of Water Temp

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1 Executive Summary

1.1 Introduction

Power plants are constrained to environmental factors. We must adapt to these constraints to maximize efficiency and avoid hazards. We focused on the James M. Barry Electric Generating Plant for this analysis.

1.2 Methods

We used historical temperature from the North Oceanic Atmospheric Administration and energy data from the U.S. Energy Information Administration. Monthly temperature averages and proportions of days exceeding 85°F were calculated by each month to define risk levels. Risk levels were defined as High being above 50% of the month having temperatures exceeding 85°F and moderate as being above 30%. Linear regression models were used to estimate projected temperature trends. Energy Generation was compared with Alabama Power Company's Energy consumption to identify impact of running at reduced capacity levels between 10% to 30%. We assossiated reduced capacity of 10% with temperature levels between 85°F and 95°F.

1.3 Results

The analysis showed seasonal increases in temperature during the months of April through October, with relevant risk from June through September. As shown in Figure 4.1, July and August are identified as high risk, while **June (45.5%)** and **September (32%)** were identified as moderate risk. These months will approach or exceed critical thresholds over the next five years (Figure 4.2). Energy generation remains sufficient across all evaluated reduced capacity scenarios. Under underperforming periods where we will not meet demands, the Alabama Power Company system can meet demand through load balancing.

1.4 Conclusion

The analysis indicates that reduced capacity is managable under current conditions. Despite summer months having increased risk, the power system is able to compensate to avoid negative impact. However, continued increase in temperature and demand can increase risks overtime. For future analyses we should incorporate additional data and broader system considerations.

> **!** Important
>
> 1. The company should run at reduced efficiency during the moderate risk and high risk months (June - September).
> 2. To ensure security and adjust effectively, the company should conduct this analysis with information on the other power plants to be prepared for temperature rising.

2 Introduction

Power plants can face operational constraints due to environmental conditions. These constraints can lead to plants being unable to meet demands and forcing decreased efficiency. One of the most impactful constraints is cooling because can cause emergency shutdown and equipment damage.A way this has been mitigated is by having a connected system to help compensate for local constraints of plants. This solution has been adopted by the Alabama Power Company, which operates 81 generating units and 26 facilities “Generating Plants” (2026).

It is unknown how often a power plant will have to run at a reduced capacity as well as the reduction requirements, while still being able to meet demands. The objective of this report is to assess the risk that the Dauphin Island power plant will have to run at reduced capacity as well as the total impact. This study analyzes the frequency of months that exceed 85°F and the generation at different reduced-capacity scenarios. This study will also project the next 5 years.

3 Methods

3.1 Water Temperature

We obtained a water temperature dataset[1] from the National Oceanic and Atmospheric Administration. The dataset had covered the hourly temperature observations from the years of 2002 to 2025 for the station in Dauphin Island “Station Home Page - NOAA Tides & Currents” (2026). There were no available observations for the year of 2021, due to Hurricane Ida. This data was used to identify the likelihood of the water temperature exceeding the temperature threshold that the plant is able to handle (> 85°F) using a confidence interval of 95%. Monthly average temperatures and the proportion of days exceeding 85°F were calculated. Months were labeled depending on proportion of the month where temperature exceeds 85°F. Months that had over 50% of observations over 85°F was labeled as *high.* Months that were in between 50% and 20% were labeled as *moderate.* We had then projected the next 5 years, and identified, which months would need to be ran at reduced capacity. To do this we had ran a linear regression model to find the trend and follow the trend for the projection.

3.2 Power Requirements

We obtained a dataset[2] on consumption from the U.S. Energy Information Administration. The dataset includes the consumption sales, per month, from all power companies from the years of 2002 to 2025 “Form EIA-861M (Formerly EIA-826) Detailed Data - u.s. Energy Information Administration (EIA)” (2026). This dataset was used to find the consumption demands required by Alabama Power Company to meet demands (Megawatt hours).

We obtained a dataset[3] on generation from the U.S. Energy Information Administration. The dataset had covered net generation, per month, that the James M. Barry Electric Generating Plant from the years of 2002 to 2025 “Electricity Data Browser - Barry” (2026). This dataset was used because Dauphin Island receives its power from the company Alabama Power Company, and this could be used to find the contribution a power plant would take “Alabama Power Service Territory” (2026).

[1]https://tidesandcurrents.noaa.gov/physocean.html?i d=8735180
[2]https://www.eia.gov/electricity/data/eia861m/
[3]https://www.eia.gov/beta/electricity/data/browser/#/plant/3?f req=M&start=202001&end=202512&ctype=linechart&ltype=p ALL-ALL.M&maptype=0&pin=&linechart=

This data was used to find the capabilities of the Dauphin Plant in the periods where we need to run at reduced capacity (Megawatt hours). We then compared this generation factored in with its contribution to consumption trends of Alabama Power Company. We defined reduced capacity as increments of 10%. These would be applied once temperature exceeds the applicable amount, the thresholds were starting at 85°F and increases by 10°F. We used the capacity type *Normal Capacity* as the real capacity used at the date. We are applying the assumption that the power plant does not need to go at a decreased capacity due to not being coastal, but was not found to be relevant because of Alabama Power Company being treated as a support system. We had predicted the impact over the next 5 years assuming that the rate of contribution stays consistent.

James M. Barry Electric Generating Plant was chosen because it produces the most out of the power plants under Alabama Power Company, to compensate for a power plant needing to contribute proportionately we increased the contribution rate of the power plant "Generating Plants" (2026).

4 Results

4.1 Water Temperature

The analysis shows a seasonal increase in water temperature from April through October. Temperatures peak in the month of July and August with mean temperatures of *86.62°F (95% CI: 86.57-86.68)* and *86.35°F (95% CI: 86.31-86.38)*. With the peaks these months were identified with high risk. The months *June (45.5%)* and *September (32%)* are at moderate-risk months. Other months represent low risk *(<20%)*. Based on Figure 4.1 the projected temperatures from June to September roughly around 85°F.

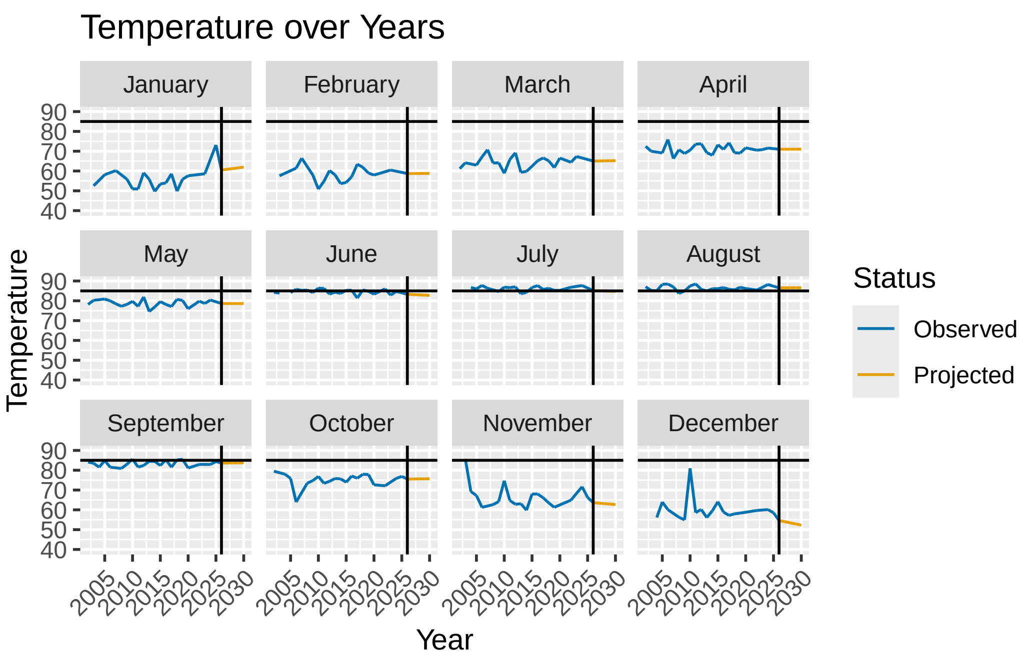

Figure 4.1: This figure shows the mean temperature throughout each year from 2002 to 2025. The line shows the threshold where a power plant would need to run at reduced risk. The projected temperature is denoted by years after the verticle line at the start 2026.

4.2 Power Requirement

Alabama Power Company produces a com bined capacity of 14 million Kilowatts (122.72 million megawatt-hours) "Generating Plants" (2026). James M. Barry Power Plant contributes to *10.3%* of the maximum capacity. Total electrical demand represents approximately 40.4% of Alabama Power Company's total capacity. James M. Barry Power Plant contributes to around *4.2%* of the overall consumption. As shown in Figure 4.2 generation remains sufficient across all reduced capacity scenarios. The highest percentage consumed is April of 2023 at *50%* of capacity, under a *30%* reduced capacity rate. The projected months did not show to have impactful rates.

The analyses shows that the impact of running at reduced capacity is minimal assuming that the plant was running at regular capacity. The highest consumed month was April, which corresponds to a low-risk period of needing to be at reduced capacity.

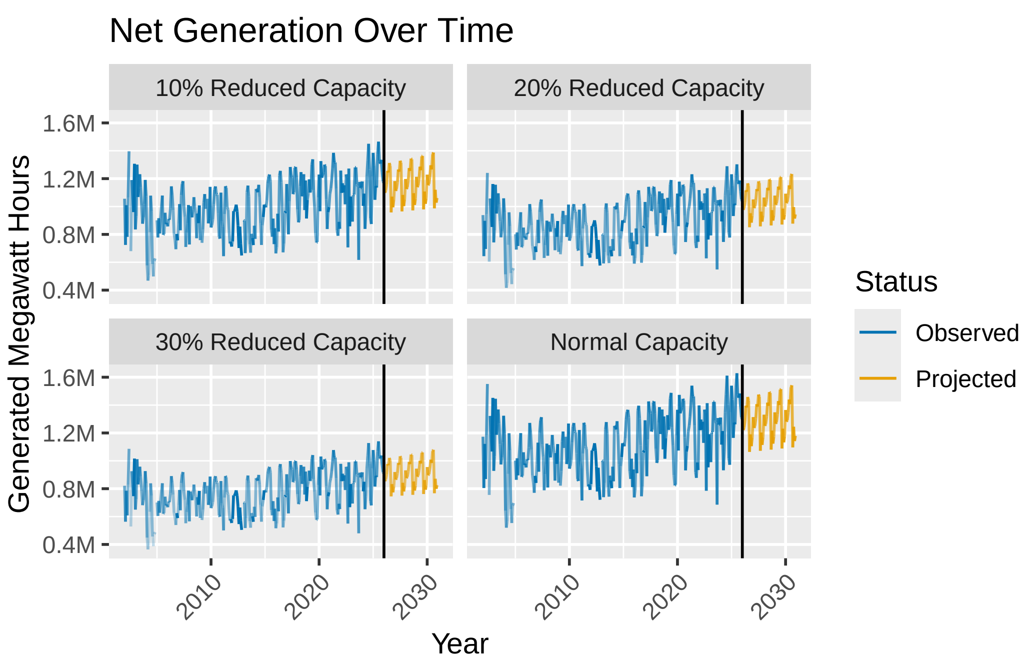

Figure 4.2: This figure shows the generation proficiency at different operational capabilities at a given time. Transparency is denoted by percentage consumed. The projected Net generation is denoted by years after the verticle line at the start 2026.

5 Discussion

5.1 Interpretation

This report analyzed the total impact of running at different levels of reduced capacity within the Dauphin Island power plant. We found what time periods of the year we would likely need to run at reduced capacity due to cooling constraints. The months that are at high risk, July and August, are months that will likely need to be run at reduced capacity of 10% over the next 5 years. Months with moderate risk, June and September, are recommended to be prepared to operate at reduced capacity. There were no instances of temperature requiring reduced capacity of 20%. Despite reduced capacity during elevated temperatures, the power plant is sufficient to meet demands. Under a scenario where reduced capacity is impactful, it can be offset due to the Alabama Power Company's load balancing.

5.2 Limitations

This study used public data from U.S. EIA and NOAA: - The data on the generation from U.S. EIA were gathered from their EIA Beta to recieve the most up-to-date data. The data may have programming errors, but was found to not be substantial considering the support network. - This study only uses the generation data for the James M. Barry Electric Generating Plant. This study does not contain data on any other Alabama Power Company Power Plant. - There was no data for water temperature prior to 2002.

5.3 Replication

While the results indicate that reduced capacity is manageable under elevated temperatures and current demands, several limitations should be acknowledged and addressed for future analyses. Reproducing the power analysis for each of the given power plants would identify real contribution percentages for each power plant. Gathering data on temperature trends would allow the company to stay informed on rising temperature patterns.

Important

1. The company should run at reduced efficiency during the moderate risk and high risk months (June - September).
2. To ensure security and adjust effectively, the company should conduct this analysis with information on the other power plants to be prepared for temperature rising.

References

"Alabama Power Service Territory." 2026. https://southerncompany.maps.arcgis.com/apps/webappviewer/index.html?id=a5e08f05ccf44e9ebae30129d8802387.

"CO-OPS Products - NOAA Tides & Currents." 2026. https://tidesandcurrents.noaa.gov/products.html.

"Electricity Data Browser - Barry." 2026. https://www.eia.gov/beta/electricity/data/browser/#/plant/3?freq=M&start=202001&end=202512&ctype=linechart&ltype=pin&tab=generation&pin=ELEC.CONS_TOT.COW-US-99.M&maptype=0&linechart=ELEC.PLANT.GEN.3-ALL-ALL.M&columnchart=ELEC.PLANT.GEN.3-ALL-ALL.M.

"FOIA: Freedom of Information Act | National Oceanic and Atmospheric Administration." 2026. https://www.noaa.gov/information-technology/foia.

"Form EIA-861M (Formerly EIA-826) Detailed Data - u.s. Energy Information Administration (EIA)." 2026. https://www.eia.gov/electricity/data/eia861m/.

"Generating Plants." 2026. https://www.alabamapower.com/company/about-us/generating-plants.html.

"Opendata - u.s. Energy Information Administration (EIA)." 2026. https://www.eia.gov/opendata/index.php.

"Optimum Interpolation Sea Surface Temperature (OISST)." 2020. https://www.ncei.noaa.gov/products/optimum-interpolation-sst.

"STAR - GOES-r Algorithm Working Group - Land and Ocean Data Products - Sea Surface Temperature." 2026. https://www.star.nesdis.noaa.gov/goesr/product_sst.php.

"Station Home Page - NOAA Tides & Currents." 2026. https://tidesandcurrents.noaa.gov/stationhome.html?id=8735180.

"Survey Forms - u.s. Energy Information Administration (EIA)." 2026. https://www.eia.gov//survey/index.php.

A Data Documentation

A.1 Water Temperature

A.1.1 Background

This dataset was collected from the National Oceanic Atmospheric Administration (NOAA) "Station Home Page - NOAA Tides & Currents" (2026). The data[1] covers hourly water temperature measurements from the years of 2020 to 2024. The data was meant to calculate the temperature trends at the Dauphin Island, AL station (ID: 8735180) "Station Home Page - NOAA Tides & Currents" (2026). The temperature data was collected using meteorological in 6 minute intervals "CO-OPS Products - NOAA Tides & Currents" (2026). There are two format type of data one in 6 minute increments and the other in hourly increments. To validate this data there are three algorithms that collect and process data and then are compared against satellite data "STAR - GOES-r Algorithm Working Group - Land and Ocean Data Products - Sea Surface Temperature" (2026). Due to the freedom of Information Act any person has the right to obtain the records "FOIA: Freedom of Information Act | National Oceanic and Atmospheric Administration" (2026). NOAA use Optimum Interpolation Sea Surface Temperature, which integrate a variety of ways that collects data and puts it in a global grid "Optimum Interpolation Sea Surface Temperature (OISST)" (2020).

A.1.2 Codebook

The main variable of interest is *Water.Temp...F.*, where it is the water temperature given at the hourly intervals. They measured the temperature in Fahrenheit, missing values are indicated with -. It is important to note that there is no data in 2021 due to Hurricane Ida as well as missing months for the following years: 2022: January, February 2023: July, August, September, October, November, December 2024: January, February, March

[1]https://tidesandcurrents.noaa.gov/physocean.html?i d=8735180

A.2 Energy Production

A.2.1 Background

Data from the James M. Barry Generating Plant was collected by the U.S. Energy Information Administration "Electricity Data Browser - Barry" (2026). The data[2] gives the net generation of energy, in megawatt hours (MWh), over the years of 2020 to 2024. The power plant reports the data to the EIA "Survey Forms - u.s. Energy Information Administration (EIA)" (2026). The data gathers the MWh generated for a given month. The data is free and open to the public "Opendata - u.s. Energy Information Administration (EIA)" (2026). The data uses megawatt hours because it gathers how many megawatts were generated for a given month.

A.2.2 Codebook

The relevant variables are columns for each documented month, where it collects the megawatt hours for that month. Missing data was indicated with – there is missing data for January of 2020 and 2023.

A.3 Energy Consumption

A.3.1 Background

Data on electricity consumption for Alabama Power Company was collected by the U.S. Energy Information Administration (EIA) "Form EIA-861M (Formerly EIA-826) Detailed Data - u.s. Energy Information Administration (EIA)" (2026). The data[3] gives the overall consumption of energy, in megawatt hours (MWh), over the years of 2020 to 2024. The data is reported from the power plant and sent to the EIA "Survey Forms - u.s. Energy Information Administration (EIA)" (2026). The data gathers the megawatt hours consumed for a given month. The data is free and open to the public "Opendata - u.s. Energy Information Administration (EIA)" (2026). The data uses megawatt hours because it gathers how many megawatts were generated for a given month.

[2] https://www.eia.gov/beta/electricity/data/browser/#/plant/3?freq=M&start=202001&end=202512&ctype=linechart&ltype=p ALL-ALL.M&maptype=0&pin=&linechart=

[3] https://www.eia.gov/electricity/data/eia861m/

A.3.2 Codebook

The variable name of interest is TotalS, which is the total sold for a given month. The data, in megawatt hours, does not contain any missing data. It is important to note that this is the consumption of Alabama Power Company not of the specific power plant, because of this we will find the contribution that the power plant makes towards the total generation.